

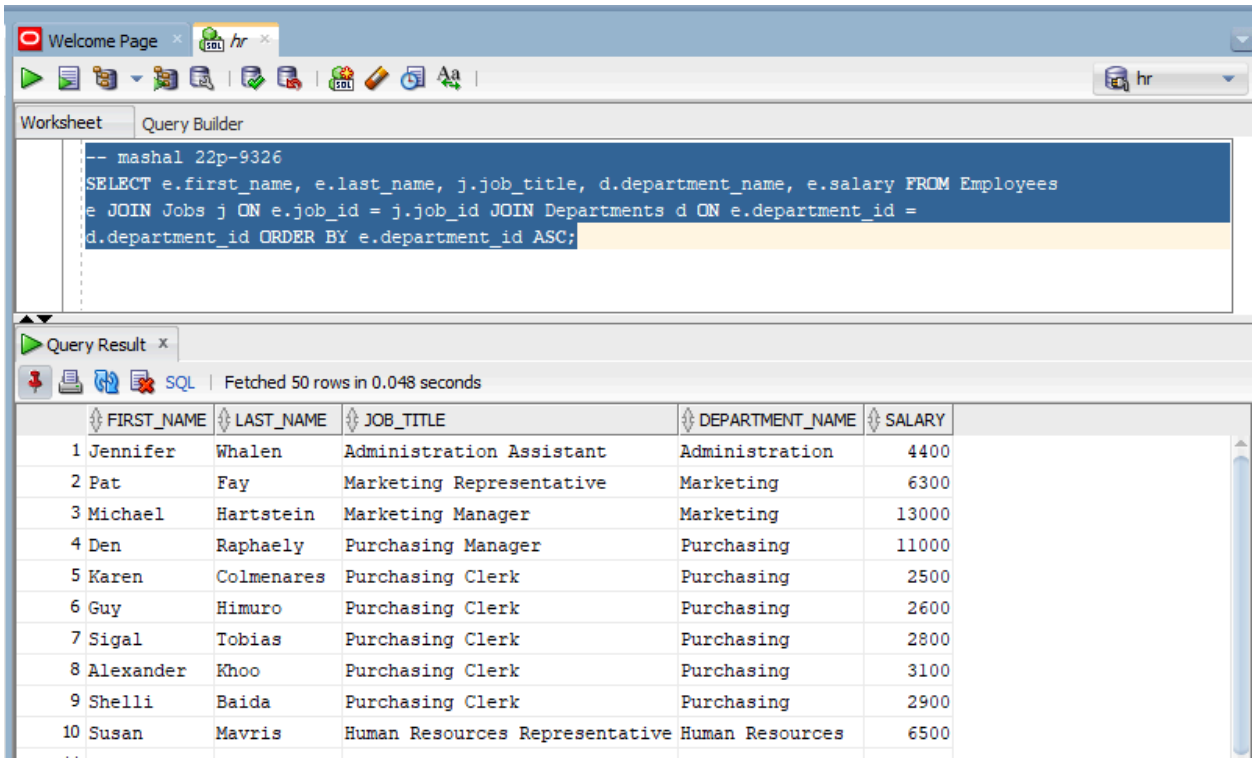
DataBase System Lab

Lab Task 5

NAME : MASHAAL ISHTIAQ
ROLL NO: 22P-9326
SECTION: BSE-6B

Query 1:

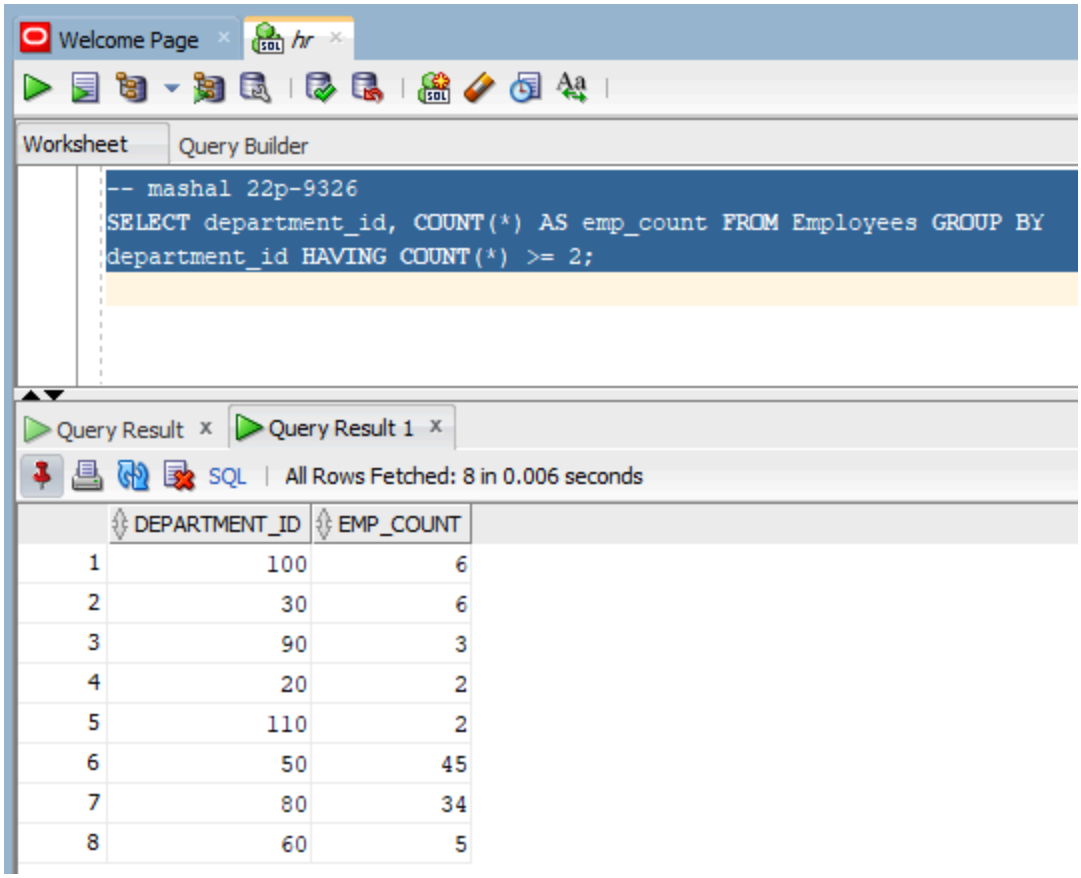
SELECT e.first_name, e.last_name, j.job_title, d.department_name, e.salary FROM Employees e JOIN Jobs j ON e.job_id = j.job_id JOIN Departments d ON e.department_id = d.department_id ORDER BY e.department_id ASC;



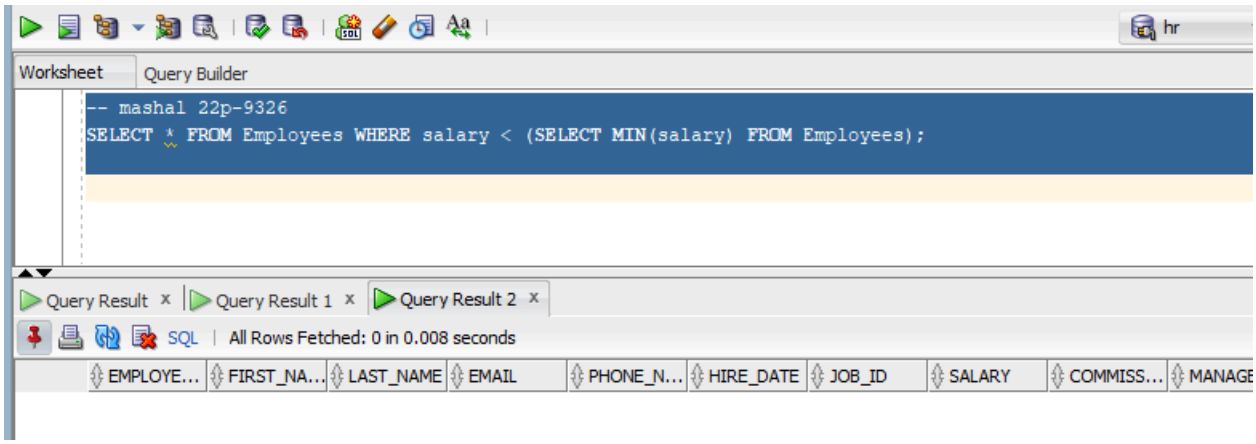
	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	SALARY
1	Jennifer	Whalen	Administration Assistant	Administration	4400
2	Pat	Fay	Marketing Representative	Marketing	6300
3	Michael	Hartstein	Marketing Manager	Marketing	13000
4	Den	Raphaely	Purchasing Manager	Purchasing	11000
5	Karen	Colmenares	Purchasing Clerk	Purchasing	2500
6	Guy	Himuro	Purchasing Clerk	Purchasing	2600
7	Sigal	Tobias	Purchasing Clerk	Purchasing	2800
8	Alexander	Khoo	Purchasing Clerk	Purchasing	3100
9	Shelli	Baida	Purchasing Clerk	Purchasing	2900
10	Susan	Mavris	Human Resources Representative	Human Resources	6500

Query 2:

SELECT department_id, COUNT(*) AS emp_count FROM Employees GROUP BY department_id HAVING COUNT(*) >= 2;

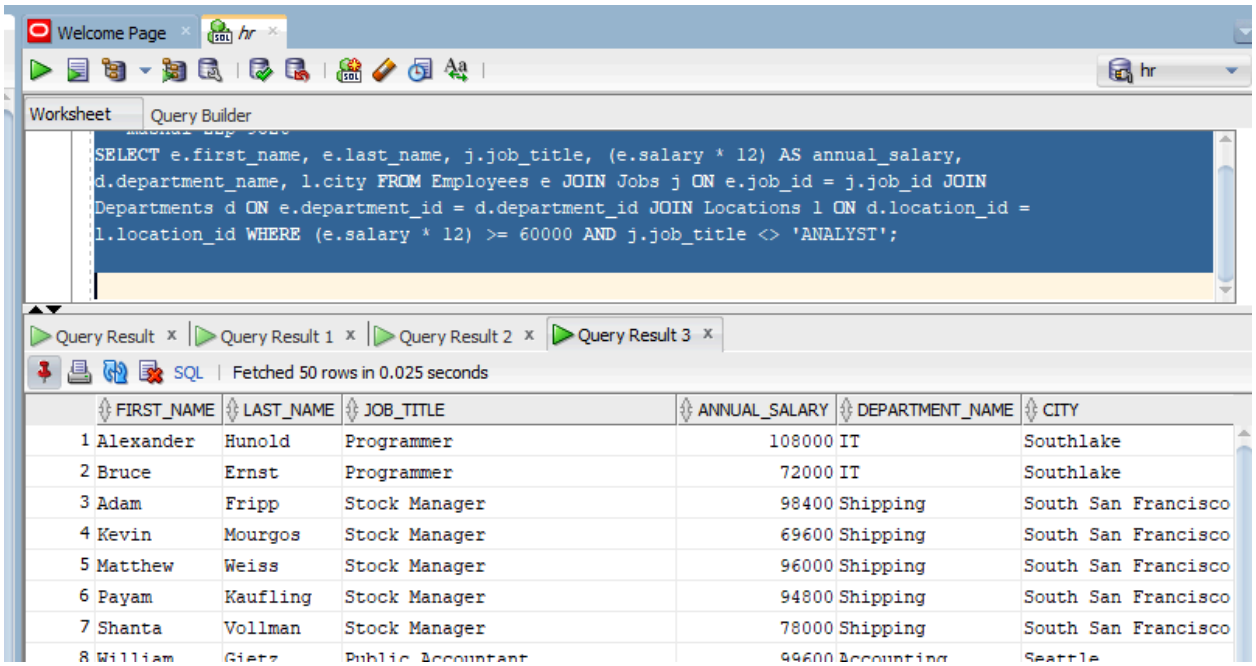


Query 3:
SELECT * FROM Employees WHERE salary < (SELECT MIN(salary) FROM Employees);

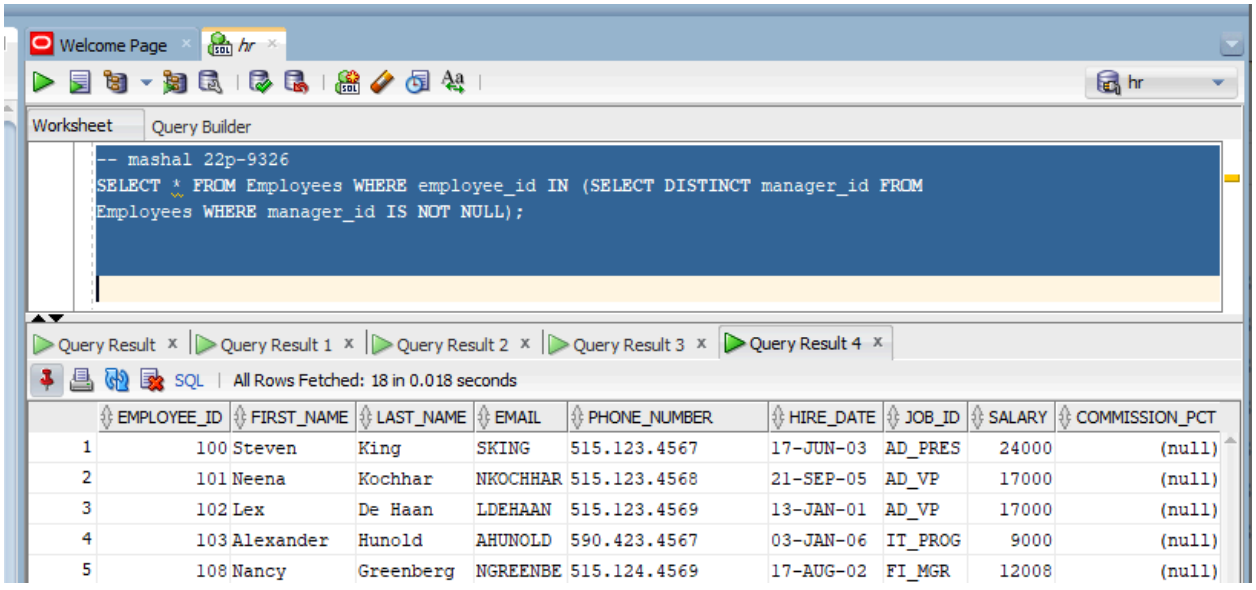


Query 4:
SELECT e.first_name, e.last_name, j.job_title, (e.salary * 12) AS annual_salary, d.department_name, l.city FROM Employees e JOIN Jobs j ON e.job_id = j.job_id JOIN

Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id WHERE (e.salary * 12) >= 60000 AND j.job_title <> 'ANALYST';

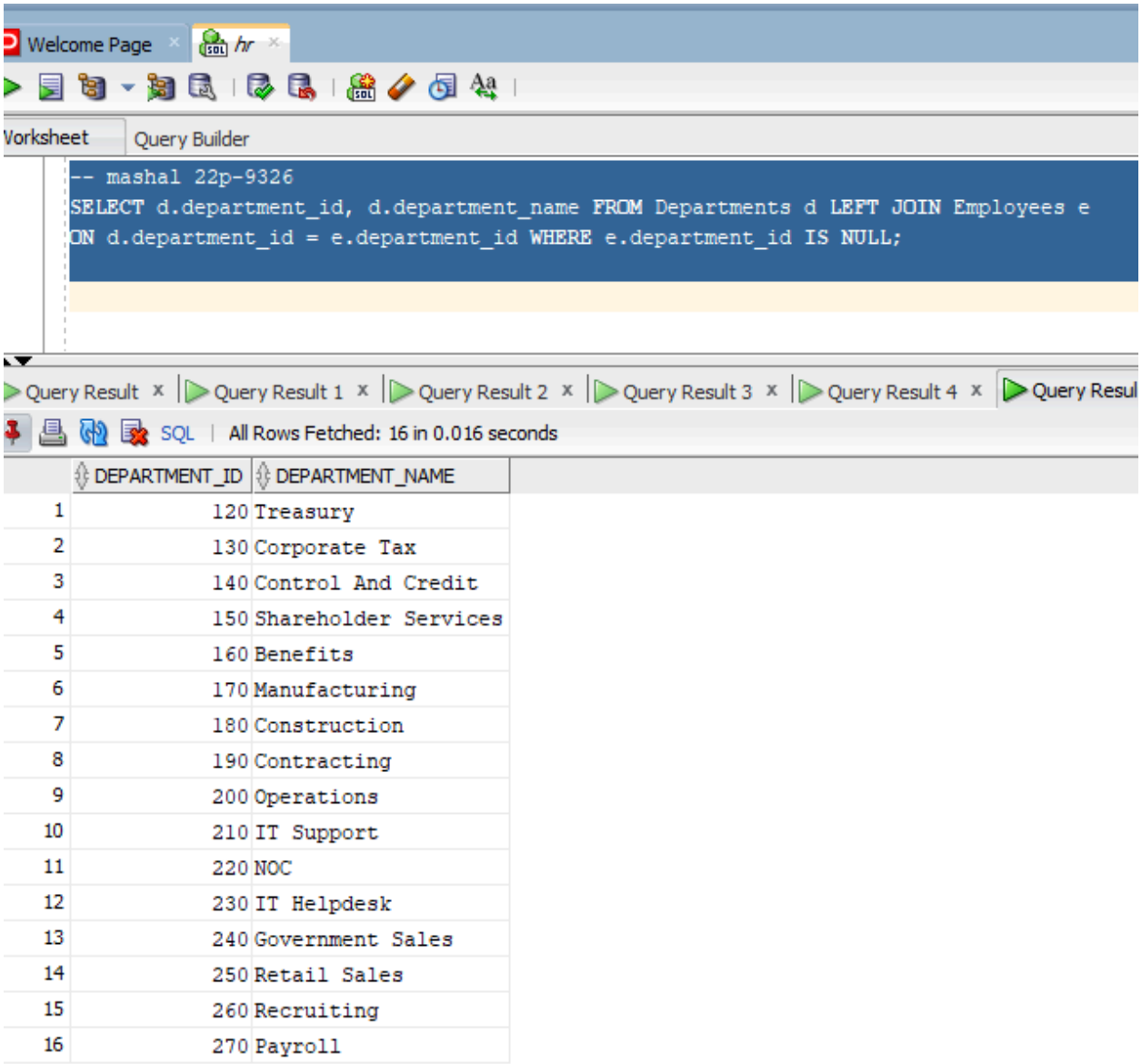


Query 5:
SELECT * FROM Employees WHERE employee_id IN (SELECT DISTINCT manager_id FROM Employees WHERE manager_id IS NOT NULL);



Query 6:

SELECT d.department_id, d.department_name FROM Departments d LEFT JOIN Employees e ON d.department_id = e.department_id WHERE e.department_id IS NULL;



	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations
10	210	IT Support
11	220	NOC
12	230	IT Helpdesk
13	240	Government Sales
14	250	Retail Sales
15	260	Recruiting
16	270	Payroll

Query 7:

SELECT e.first_name, e.last_name, e.salary, d.department_name FROM Employees e LEFT JOIN Departments d ON e.department_id = d.department_id;

ksheet | Query Builder

```
-- mashal 22p-9326
SELECT e.first_name, e.last_name, e.salary,
d.department_name FROM Employees e LEFT
JOIN Departments d ON e.department_id = d.department_id;
```

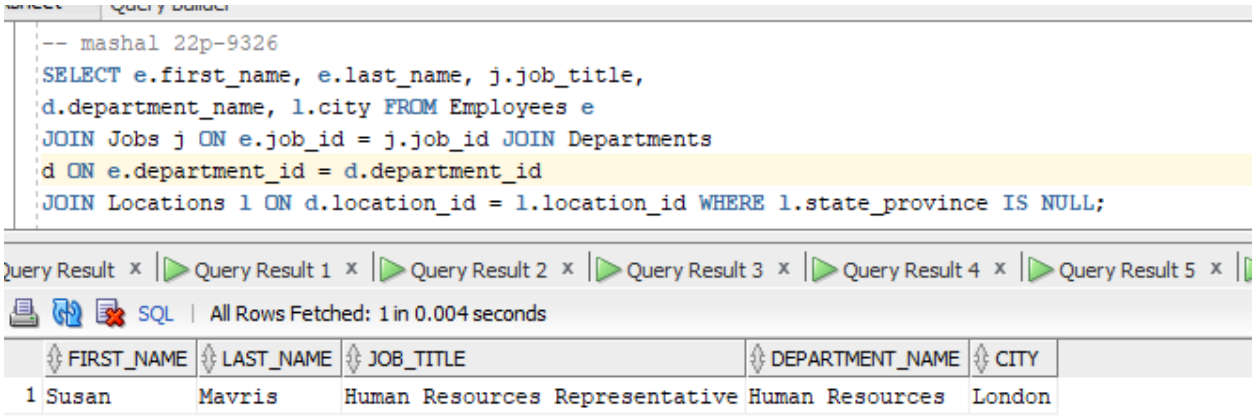
Query Result x | Query Result 1 x | Query Result 2 x | Query Result 3

SQL | Fetched 50 rows in 0.012 seconds

	FIRST_NAME	LAST_NAME	SALARY	DEPARTMENT_NAME
1	Jennifer	Whalen	4400	Administration
2	Pat	Fay	6300	Marketing
3	Michael	Hartstein	13000	Marketing
4	Karen	Colmenares	2500	Purchasing
5	Guy	Himuro	2600	Purchasing
6	Sigal	Tobias	2800	Purchasing
7	Shelli	Baida	2900	Purchasing
8	Alexander	Khoo	3100	Purchasing
9	Den	Raphaely	11000	Purchasing
10	Susan	Mavris	6500	Human Resources
11	Douglas	Grant	2600	Shipping
12	Donald	OConnell	2600	Shipping
13	Kevin	Feeney	3000	Shipping
14	Alana	Walsh	3100	Shipping
15	Vance	Jones	2800	Shipping
16	Samuel	McCain	3200	Shipping
17	Britney	Everett	3900	Shipping
18	Sarah	Bell	4000	Shipping
19	Randall	Perkins	2500	Shipping
20	Timothy	Gates	2900	Shipping

Query 8:

SELECT e.first_name, e.last_name, j.job_title, d.department_name, l.city FROM Employees e JOIN Jobs j ON e.job_id = j.job_id JOIN Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id WHERE l.state_province IS NULL;



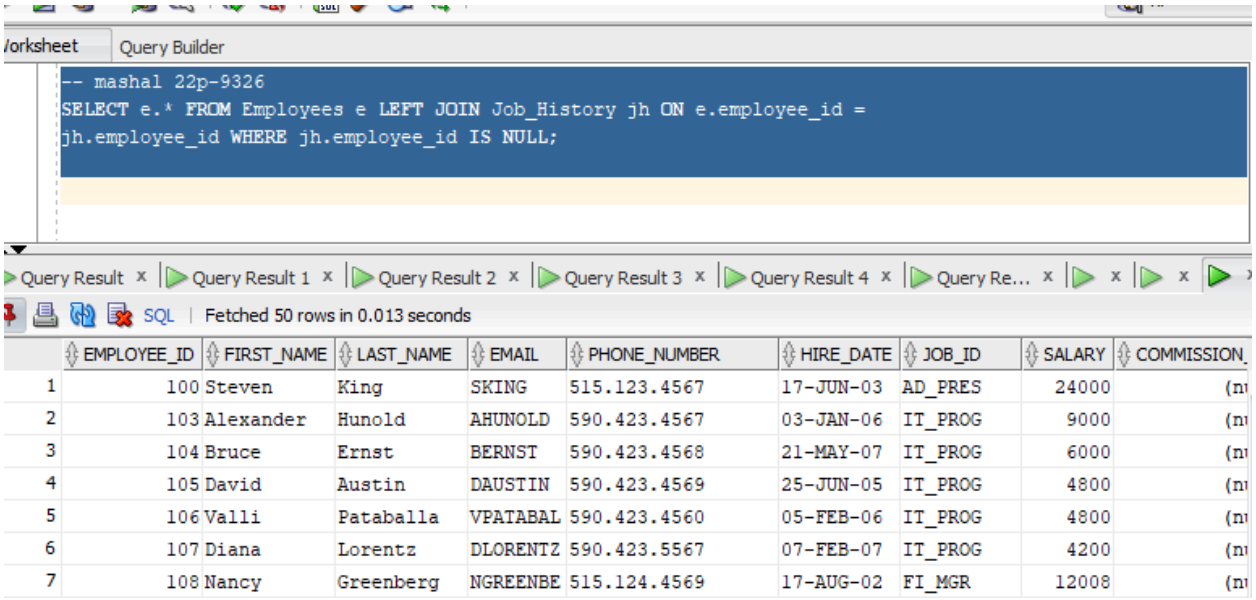
```
-- mashal 22p-9326
SELECT e.first_name, e.last_name, j.job_title,
d.department_name, l.city FROM Employees e
JOIN Jobs j ON e.job_id = j.job_id JOIN Departments
d ON e.department_id = d.department_id
JOIN Locations l ON d.location_id = l.location_id WHERE l.state_province IS NULL;
```

Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Result 5

SQL | All Rows Fetched: 1 in 0.004 seconds

	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY
1	Susan	Mavris	Human Resources Representative	Human Resources	London

Query 9:
SELECT e.* FROM Employees e LEFT JOIN Job_History jh ON e.employee_id = jh.employee_id WHERE jh.employee_id IS NULL;



Worksheet | Query Builder

```
-- mashal 22p-9326
SELECT e.* FROM Employees e LEFT JOIN Job_History jh ON e.employee_id =
jh.employee_id WHERE jh.employee_id IS NULL;
```

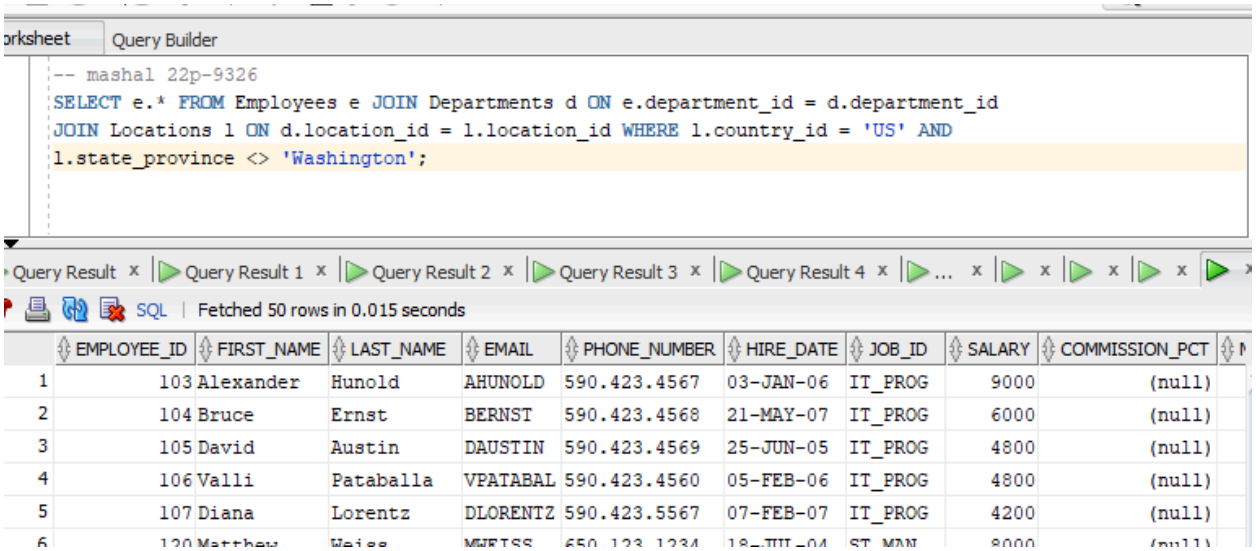
Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | Query Re...

SQL | Fetched 50 rows in 0.013 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(n
2	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(n
3	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(n
4	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(n
5	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(n
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(n
7	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(n

Query 10:

SELECT e.* FROM Employees e JOIN Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id WHERE l.country_id = 'US' AND l.state_province <> 'Washington';



orksheet | Query Builder

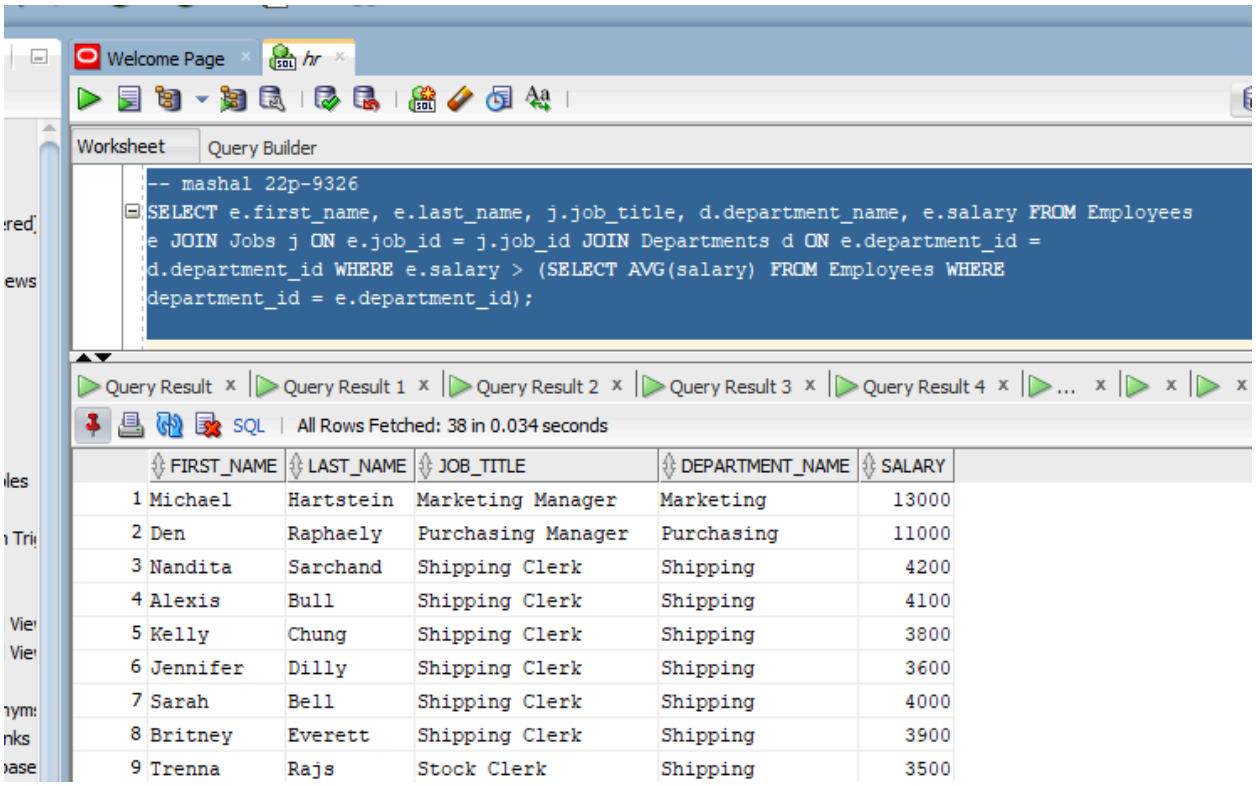
```
-- mashal 22p-9326
SELECT e.* FROM Employees e JOIN Departments d ON e.department_id = d.department_id
JOIN Locations l ON d.location_id = l.location_id WHERE l.country_id = 'US' AND
l.state_province <> 'Washington';
```

Query Result | Query Result 1 | Query Result 2 | Query Result 3 | Query Result 4 | ...

SQL | Fetched 50 rows in 0.015 seconds

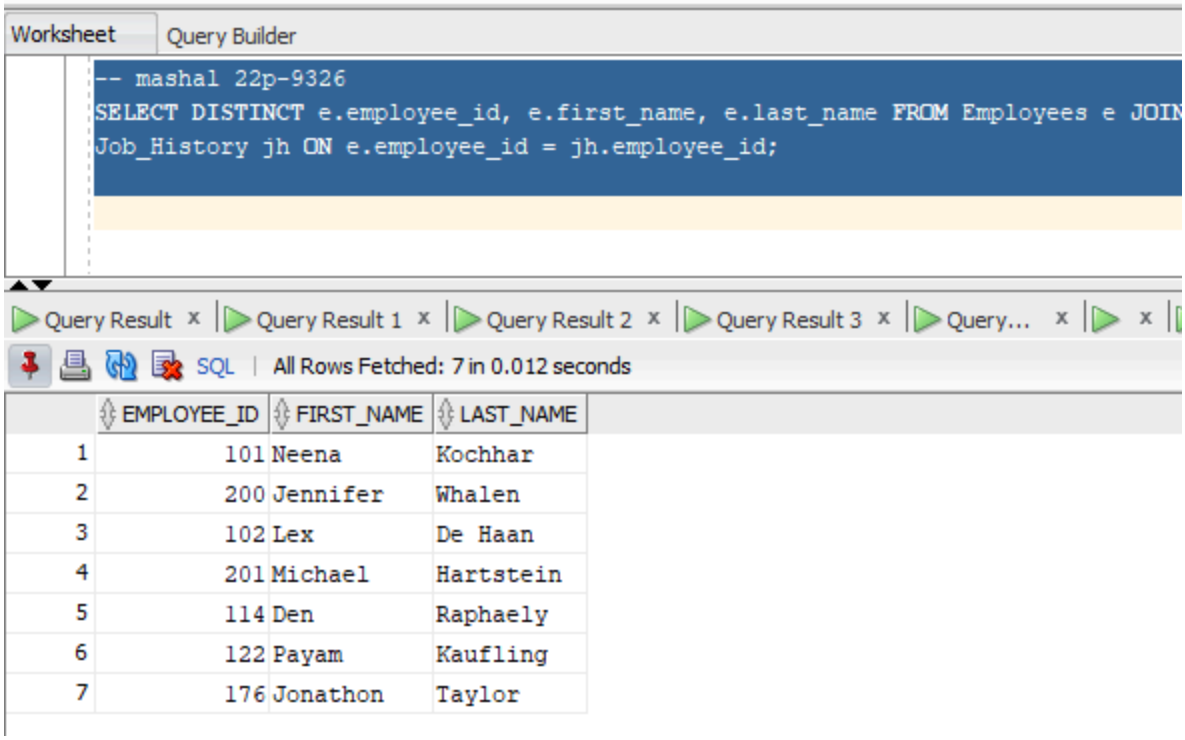
	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
1	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)
2	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)
3	105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG	4800	(null)
4	106	Valli	Pataballa	VPATABAL	590.423.4560	05-FEB-06	IT_PROG	4800	(null)
5	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)
6	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(null)

Query 11:
SELECT e.first_name, e.last_name, j.job_title, d.department_name, e.salary FROM Employees e JOIN Jobs j ON e.job_id = j.job_id JOIN Departments d ON e.department_id = d.department_id WHERE e.salary > (SELECT AVG(salary) FROM Employees WHERE department_id = e.department_id);



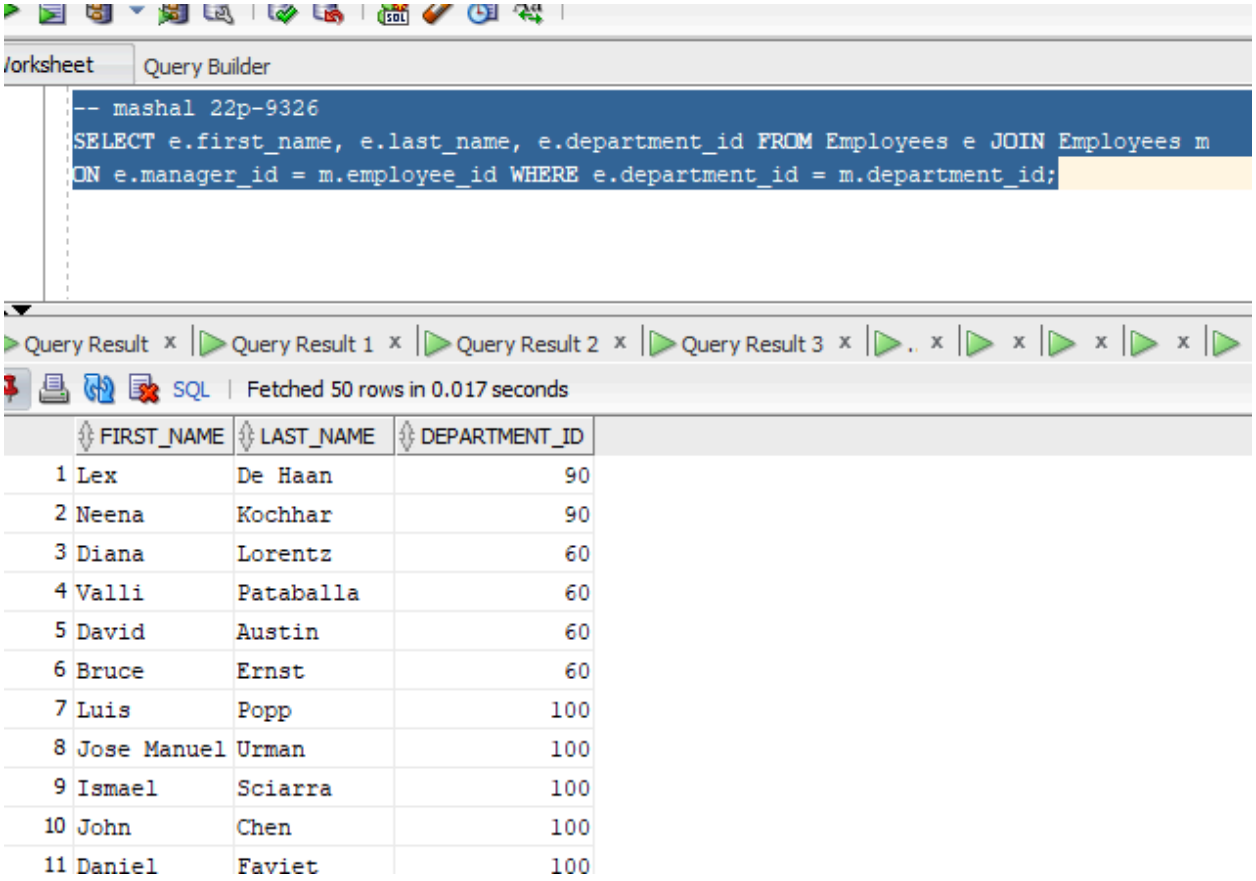
	FIRST_NAME	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	SALARY
1	Michael	Hartstein	Marketing Manager	Marketing	13000
2	Den	Raphaely	Purchasing Manager	Purchasing	11000
3	Nandita	Sarchand	Shipping Clerk	Shipping	4200
4	Alexis	Bull	Shipping Clerk	Shipping	4100
5	Kelly	Chung	Shipping Clerk	Shipping	3800
6	Jennifer	Dilly	Shipping Clerk	Shipping	3600
7	Sarah	Bell	Shipping Clerk	Shipping	4000
8	Britney	Everett	Shipping Clerk	Shipping	3900
9	Trenna	Rajs	Stock Clerk	Shipping	3500

Query 12:
SELECT DISTINCT e.employee_id, e.first_name, e.last_name FROM Employees e JOIN Job_History jh ON e.employee_id = jh.employee_id;



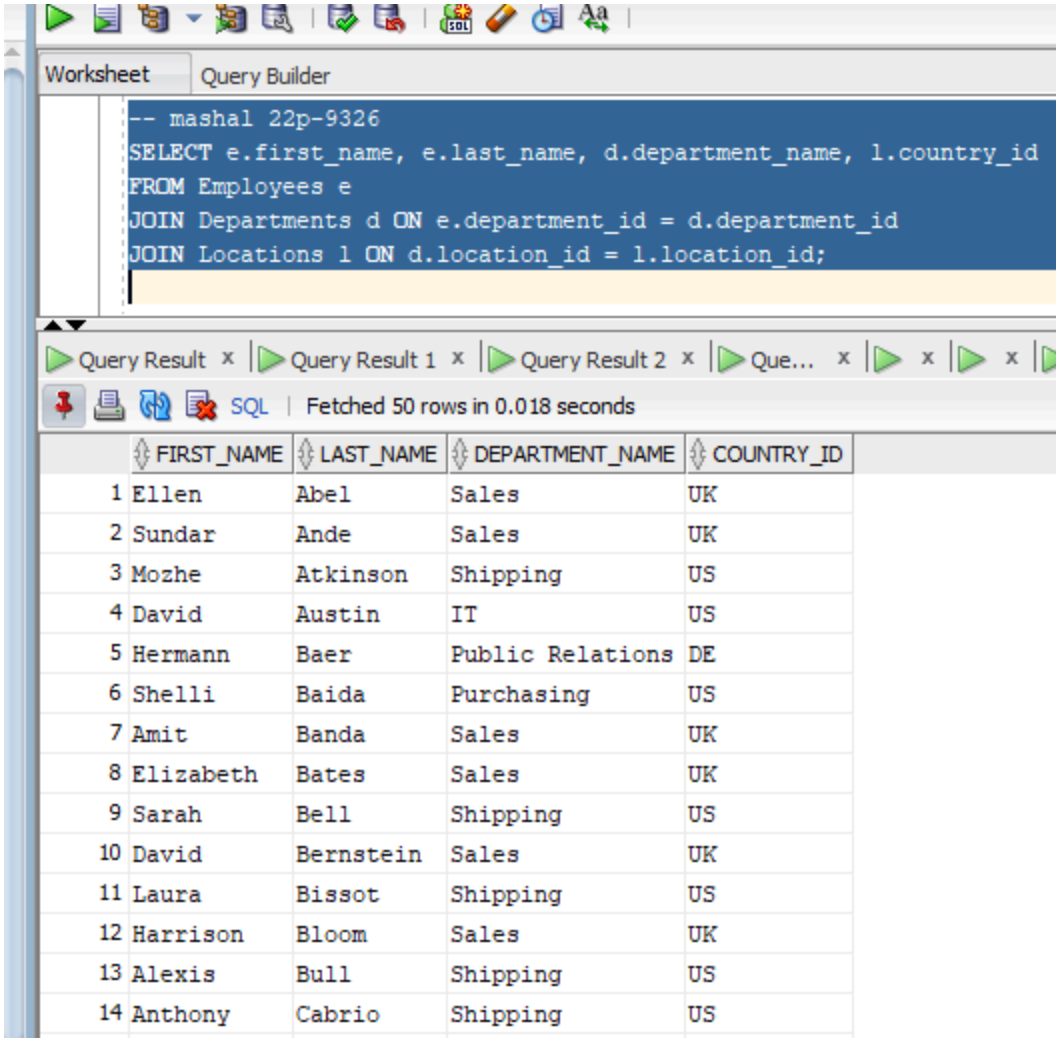
	EMPLOYEE_ID	FIRST_NAME	LAST_NAME
1	101	Neena	Kochhar
2	200	Jennifer	Whalen
3	102	Lex	De Haan
4	201	Michael	Hartstein
5	114	Den	Raphaely
6	122	Payam	Kaufling
7	176	Jonathon	Taylor

Query 13:
SELECT e.first_name, e.last_name, e.department_id FROM Employees e JOIN Employees m ON e.manager_id = m.employee_id WHERE e.department_id = m.department_id;



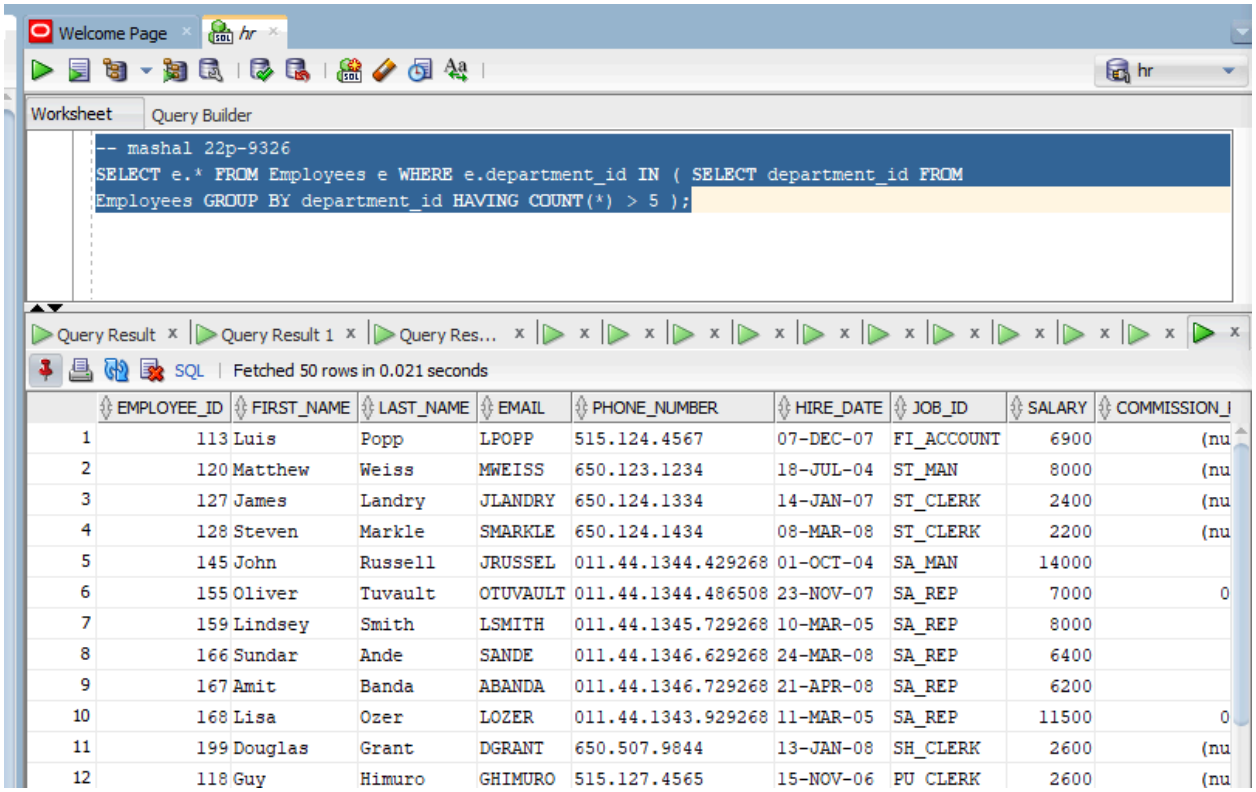
	FIRST_NAME	LAST_NAME	DEPARTMENT_ID
1	Lex	De Haan	90
2	Neena	Kochhar	90
3	Diana	Lorentz	60
4	Valli	Pataballa	60
5	David	Austin	60
6	Bruce	Ernst	60
7	Luis	Popp	100
8	Jose Manuel	Urman	100
9	Ismael	Sciarra	100
10	John	Chen	100
11	Daniel	Faviet	100

Query 14:
SELECT e.first_name, e.last_name, d.department_name, l.country_id FROM Employees e JOIN Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id;



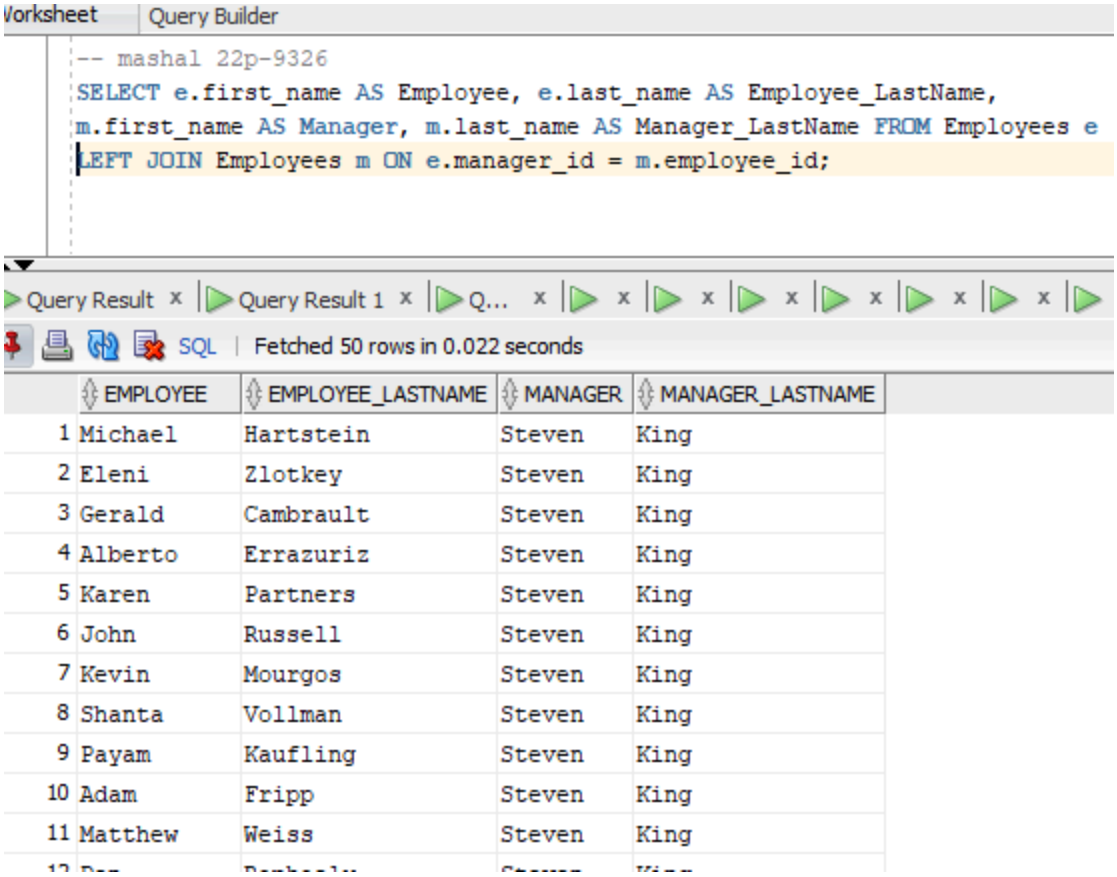
	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	COUNTRY_ID
1	Ellen	Abel	Sales	UK
2	Sundar	Ande	Sales	UK
3	Mozhe	Atkinson	Shipping	US
4	David	Austin	IT	US
5	Hermann	Baer	Public Relations	DE
6	Shelli	Baida	Purchasing	US
7	Amit	Banda	Sales	UK
8	Elizabeth	Bates	Sales	UK
9	Sarah	Bell	Shipping	US
10	David	Bernstein	Sales	UK
11	Laura	Bissot	Shipping	US
12	Harrison	Bloom	Sales	UK
13	Alexis	Bull	Shipping	US
14	Anthony	Cabrio	Shipping	US

Query 15:
SELECT e.* FROM Employees e WHERE e.department_id IN (SELECT department_id FROM Employees GROUP BY department_id HAVING COUNT(*) > 5);

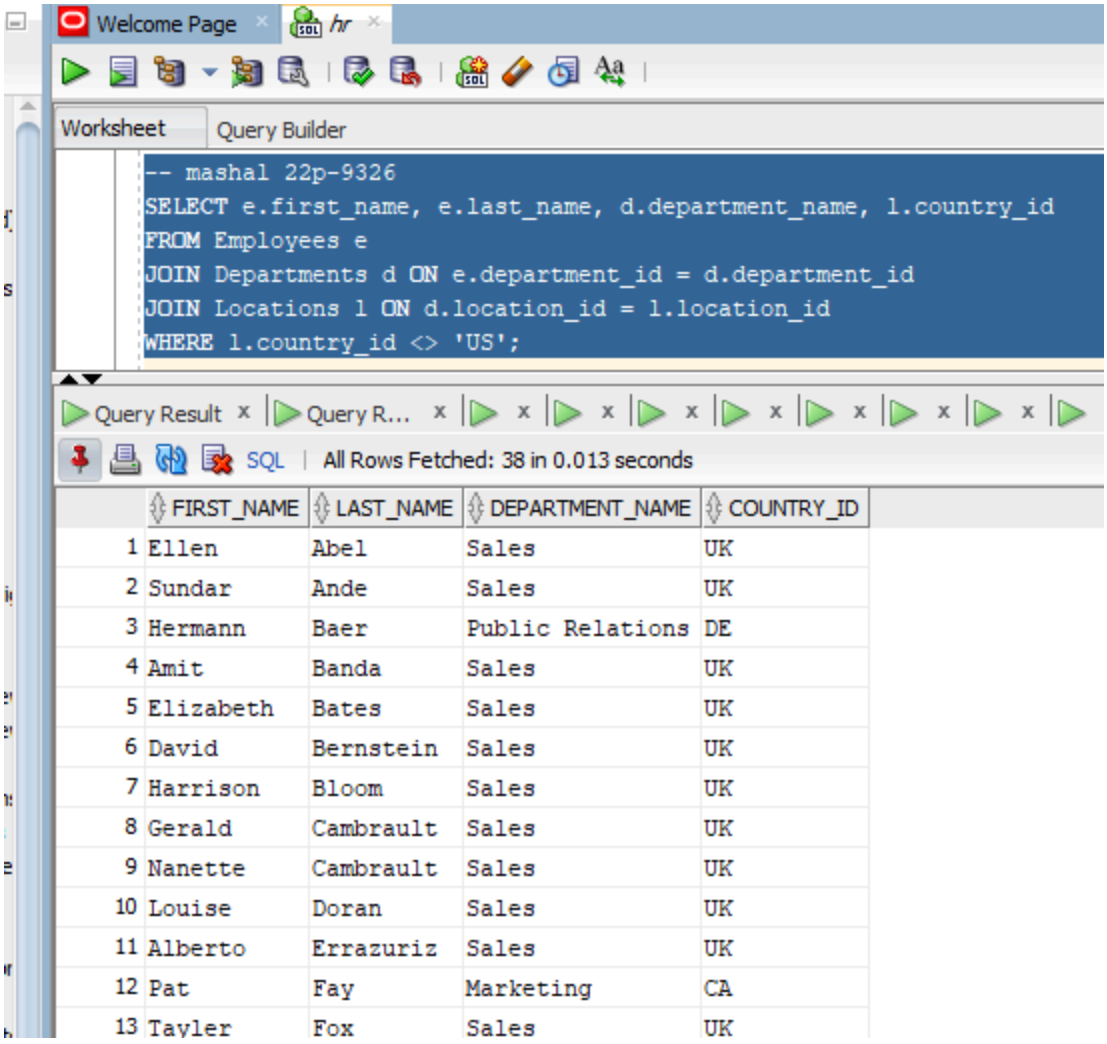


	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_I
1	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(nu
2	120	Matthew	Weiss	MWEISS	650.123.1234	18-JUL-04	ST_MAN	8000	(nu
3	127	James	Landry	JLANDRY	650.124.1334	14-JAN-07	ST_CLERK	2400	(nu
4	128	Steven	Markle	SMARKLE	650.124.1434	08-MAR-08	ST_CLERK	2200	(nu
5	145	John	Russell	JRUSSEL	011.44.1344.429268	01-OCT-04	SA_MAN	14000	
6	155	Oliver	Tuvault	OTUVAULT	011.44.1344.486508	23-NOV-07	SA_REP	7000	0
7	159	Lindsey	Smith	LSMITH	011.44.1345.729268	10-MAR-05	SA_REP	8000	
8	166	Sundar	Ande	SANDE	011.44.1346.629268	24-MAR-08	SA_REP	6400	
9	167	Amit	Banda	ABANDA	011.44.1346.729268	21-APR-08	SA_REP	6200	
10	168	Lisa	Ozer	LOZER	011.44.1343.929268	11-MAR-05	SA_REP	11500	0
11	199	Douglas	Grant	DGRANT	650.507.9844	13-JAN-08	SH_CLERK	2600	(nu
12	118	Guy	Himuro	GHIMURO	515.127.4565	15-NOV-06	PU_CLERK	2600	(nu

Query 16:
SELECT e.first_name AS Employee, e.last_name AS Employee_LastName, m.first_name AS Manager, m.last_name AS Manager_LastName FROM Employees e LEFT JOIN Employees m ON e.manager_id = m.employee_id;



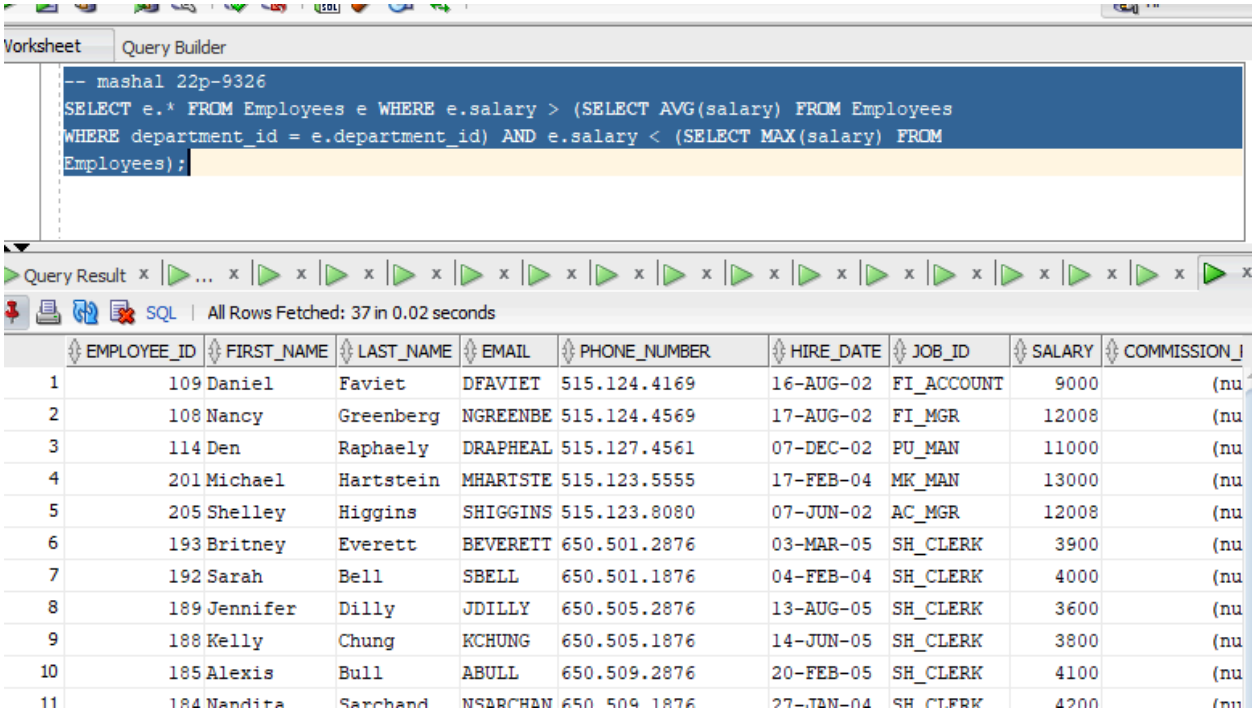
Query 17:
SELECT e.first_name, e.last_name, d.department_name, l.country_id FROM Employees e JOIN Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id WHERE l.country_id <> 'US';



	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME	COUNTRY_ID
1	Ellen	Abel	Sales	UK
2	Sundar	Ande	Sales	UK
3	Hermann	Baer	Public Relations	DE
4	Amit	Banda	Sales	UK
5	Elizabeth	Bates	Sales	UK
6	David	Bernstein	Sales	UK
7	Harrison	Bloom	Sales	UK
8	Gerald	Cambrault	Sales	UK
9	Nanette	Cambrault	Sales	UK
10	Louise	Doran	Sales	UK
11	Alberto	Errazuriz	Sales	UK
12	Pat	Fay	Marketing	CA
13	Tayler	Fox	Sales	UK

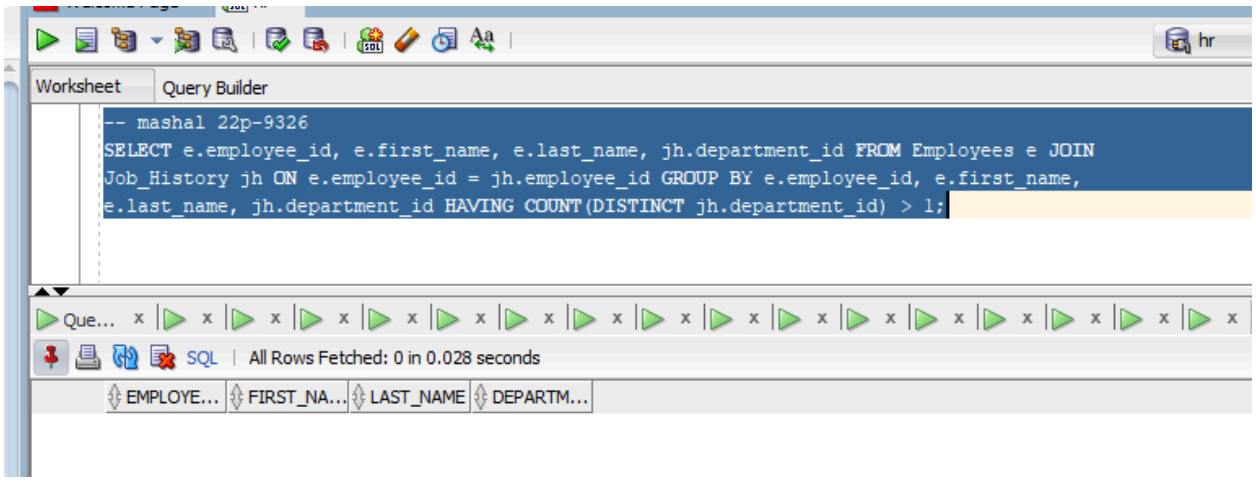
Query 18:

SELECT e.* FROM Employees e WHERE e.salary > (SELECT AVG(salary) FROM Employees WHERE department_id = e.department_id) AND e.salary < (SELECT MAX(salary) FROM Employees);

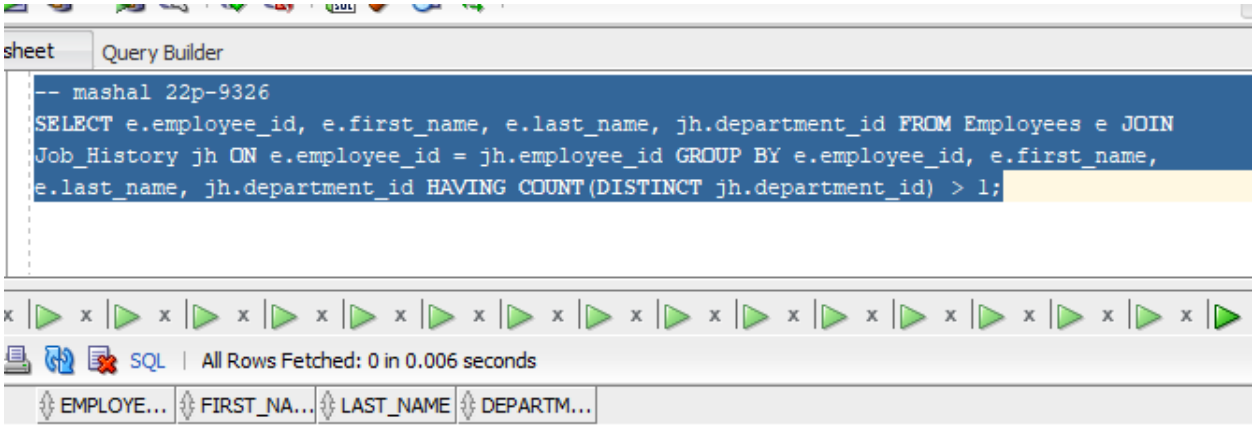


	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_I
1	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(nu
2	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(nu
3	114	Den	Raphaely	DRAPHEAL	515.127.4561	07-DEC-02	PU_MAN	11000	(nu
4	201	Michael	Hartstein	MHARTSTE	515.123.5555	17-FEB-04	MK_MAN	13000	(nu
5	205	Shelley	Higgins	SHIGGINS	515.123.8080	07-JUN-02	AC_MGR	12008	(nu
6	193	Britney	Everett	BEVERETT	650.501.2876	03-MAR-05	SH_CLERK	3900	(nu
7	192	Sarah	Bell	SBELL	650.501.1876	04-FEB-04	SH_CLERK	4000	(nu
8	189	Jennifer	Dilly	JDILLY	650.505.2876	13-AUG-05	SH_CLERK	3600	(nu
9	188	Kelly	Chung	KCHUNG	650.505.1876	14-JUN-05	SH_CLERK	3800	(nu
10	185	Alexis	Bull	ABULL	650.509.2876	20-FEB-05	SH_CLERK	4100	(nu
11	184	Nandita	Sarchand	NSARCHAN	650.509.1876	27-JAN-04	SH_CLERK	4200	(nu

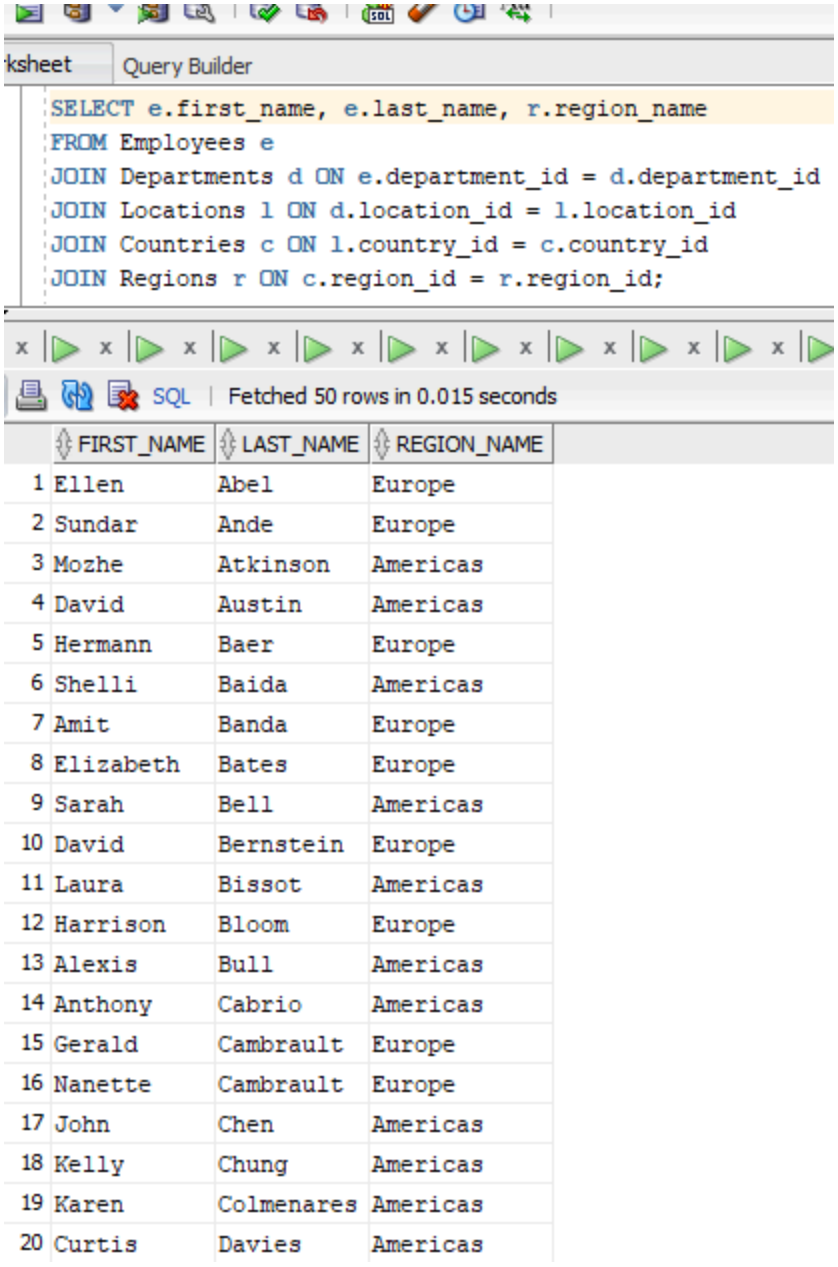
Query 19:
SELECT e.employee_id, e.first_name, e.last_name, jh.department_id FROM Employees e JOIN Job_History jh ON e.employee_id = jh.employee_id GROUP BY e.employee_id, e.first_name, e.last_name, jh.department_id HAVING COUNT(DISTINCT jh.department_id) > 1;



Query 20:
SELECT e.employee_id, e.first_name, e.last_name, jh.department_id FROM Employees e JOIN Job_History jh ON e.employee_id = jh.employee_id GROUP BY e.employee_id, e.first_name, e.last_name, jh.department_id HAVING COUNT(DISTINCT jh.department_id) > 1;



Query 21:
SELECT e.first_name, e.last_name, r.region_name FROM Employees e JOIN Departments d ON e.department_id = d.department_id JOIN Locations l ON d.location_id = l.location_id JOIN Countries c ON l.country_id = c.country_id JOIN Regions r ON c.region_id = r.region_id;



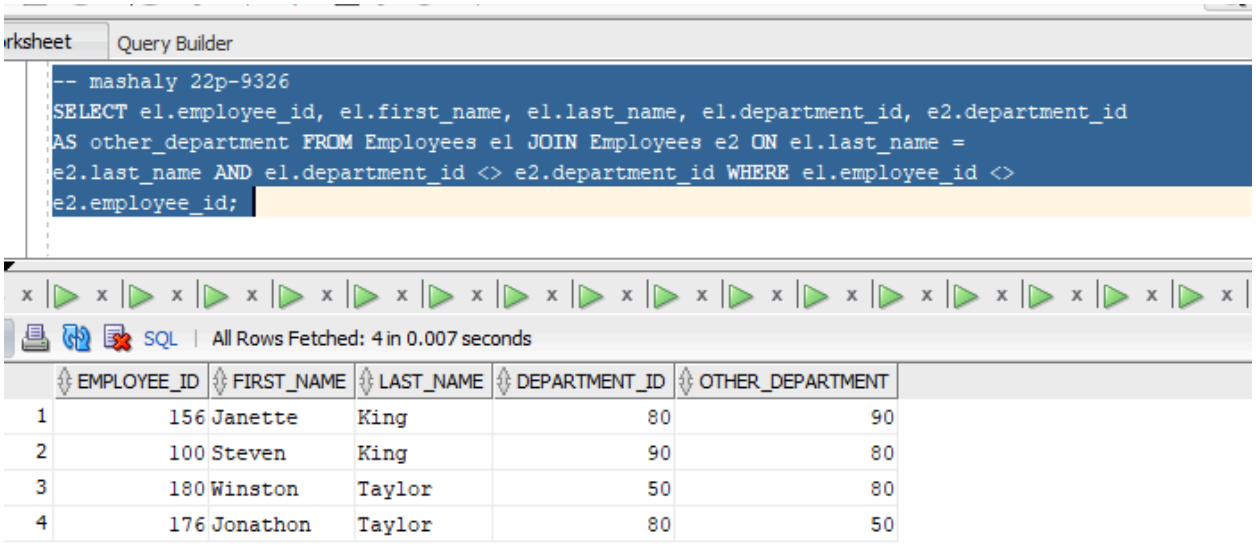
ksheet | Query Builder

```
SELECT e.first_name, e.last_name, r.region_name
FROM Employees e
JOIN Departments d ON e.department_id = d.department_id
JOIN Locations l ON d.location_id = l.location_id
JOIN Countries c ON l.country_id = c.country_id
JOIN Regions r ON c.region_id = r.region_id;
```

SQL | Fetched 50 rows in 0.015 seconds

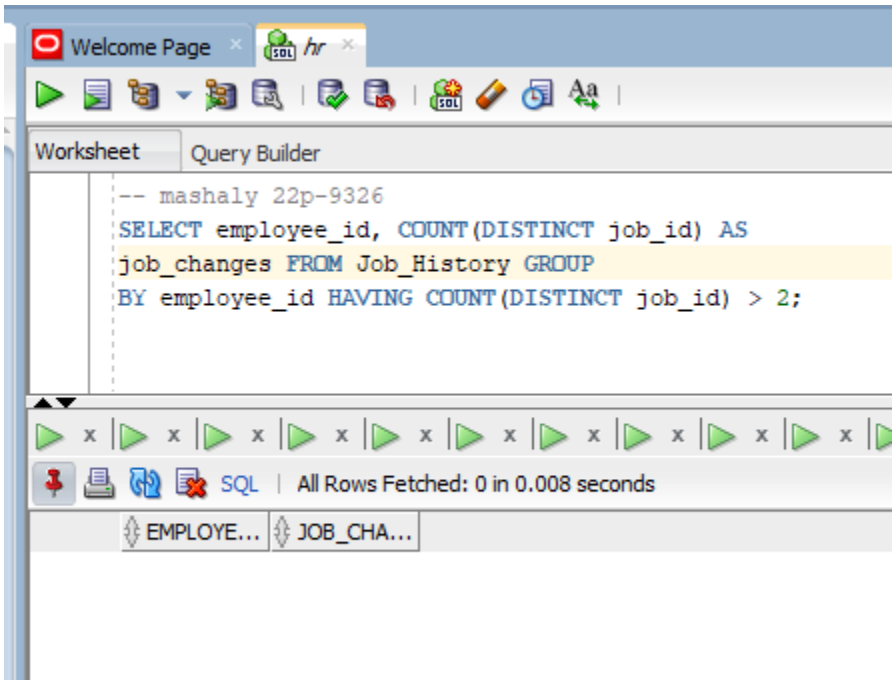
	FIRST_NAME	LAST_NAME	REGION_NAME
1	Ellen	Abel	Europe
2	Sundar	Ande	Europe
3	Mozhe	Atkinson	Americas
4	David	Austin	Americas
5	Hermann	Baer	Europe
6	Shelli	Baida	Americas
7	Amit	Banda	Europe
8	Elizabeth	Bates	Europe
9	Sarah	Bell	Americas
10	David	Bernstein	Europe
11	Laura	Bissot	Americas
12	Harrison	Bloom	Europe
13	Alexis	Bull	Americas
14	Anthony	Cabrio	Americas
15	Gerald	Cambrault	Europe
16	Nanette	Cambrault	Europe
17	John	Chen	Americas
18	Kelly	Chung	Americas
19	Karen	Colmenares	Americas
20	Curtis	Davies	Americas

Query 22:
SELECT e1.employee_id, e1.first_name, e1.last_name, e1.department_id, e2.department_id AS other_department FROM Employees e1 JOIN Employees e2 ON e1.last_name = e2.last_name AND e1.department_id <> e2.department_id WHERE e1.employee_id <> e2.employee_id;

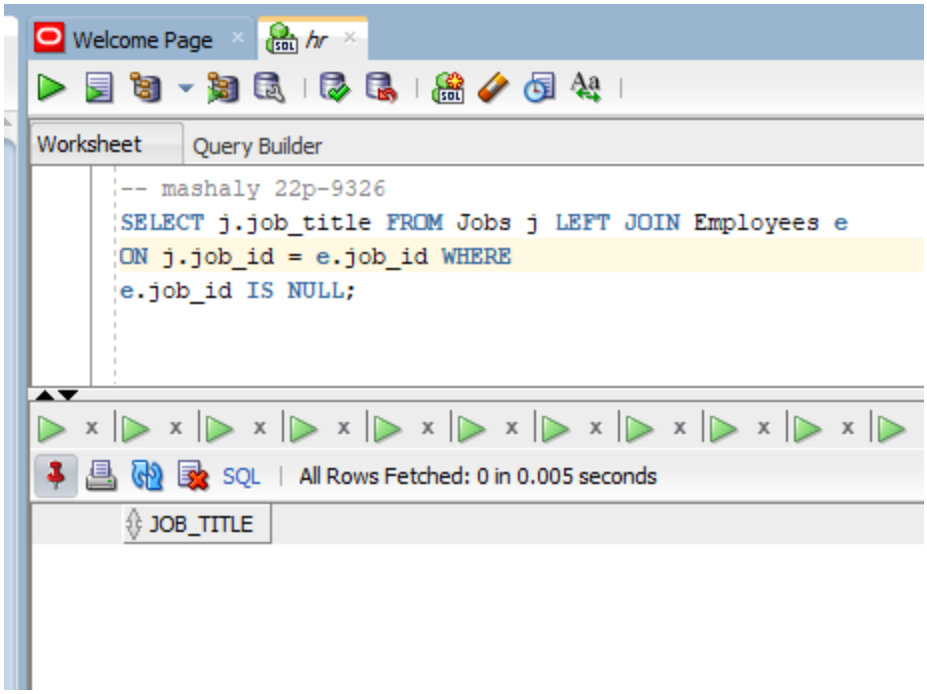


	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	OTHER_DEPARTMENT
1	156	Janette	King	80	90
2	100	Steven	King	90	80
3	180	Winston	Taylor	50	80
4	176	Jonathon	Taylor	80	50

Query 23:
SELECT employee_id, COUNT(DISTINCT job_id) AS job_changes FROM Job_History GROUP BY employee_id HAVING COUNT(DISTINCT job_id) > 2;



Query 24:
SELECT j.job_title FROM Jobs j LEFT JOIN Employees e ON j.job_id = e.job_id WHERE e.job_id IS NULL;



Query 25:
SELECT employee_id, first_name, last_name, department_id, salary FROM (SELECT e.*, RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS salary_rank FROM Employees e) WHERE salary_rank <= 3;

Worksheet | Query Builder

```
-- mashaly 22p-9326
SELECT employee_id, first_name, last_name, department_id, salary FROM ( SELECT e.*,
RANK() OVER (PARTITION BY department_id ORDER BY salary DESC) AS salary_rank
FROM Employees e ) WHERE salary_rank <= 3;
```

SQL | All Rows Fetched: 27 in 0.008 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	DEPARTMENT_ID	SALARY
1	200	Jennifer	Whalen	10	4400
2	201	Michael	Hartstein	20	13000
3	202	Pat	Fay	20	6300
4	114	Den	Raphaely	30	11000
5	115	Alexander	Khoo	30	3100
6	116	Shelli	Baida	30	2900
7	203	Susan	Mavris	40	6500
8	121	Adam	Fripp	50	8200
9	120	Matthew	Weiss	50	8000